SYSTEMS MANAGEMENT



From Lab to Fleet:

A Secure, Unified Debug Framework for Cloud Systems

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OCP PLATINUM® Member

The OCP connection: the BMC on an OCP® DC-SCM™ becomes the secure debug gateway

Standard: OCP® Hardware Management • System Debug Workstream (System Debug Specification v0.1)

Platform: OCP® DC-SCM™ + OpenBMC® **Transport:** MIPI® SPP™ / Debug over I3C® **Security:** DMTF® SPDM®

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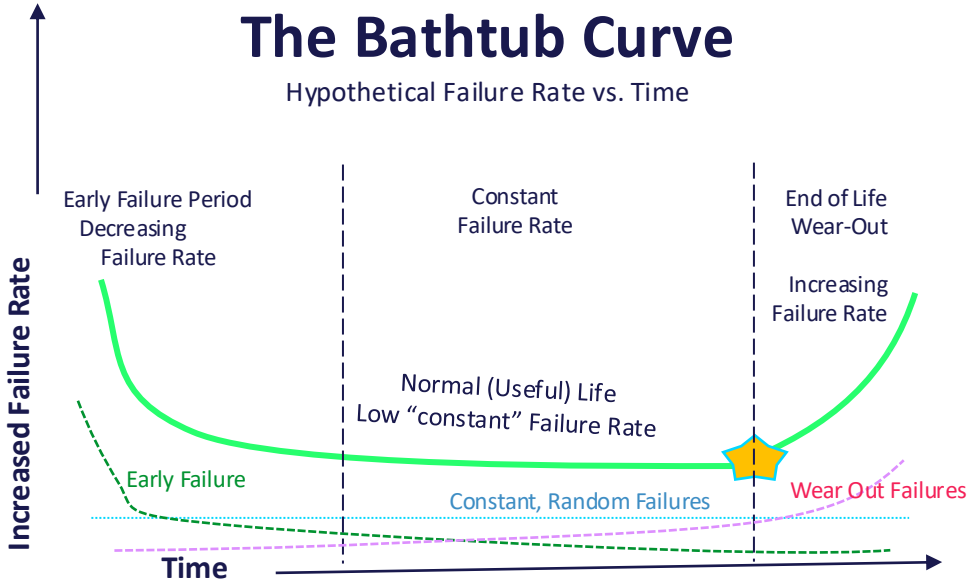
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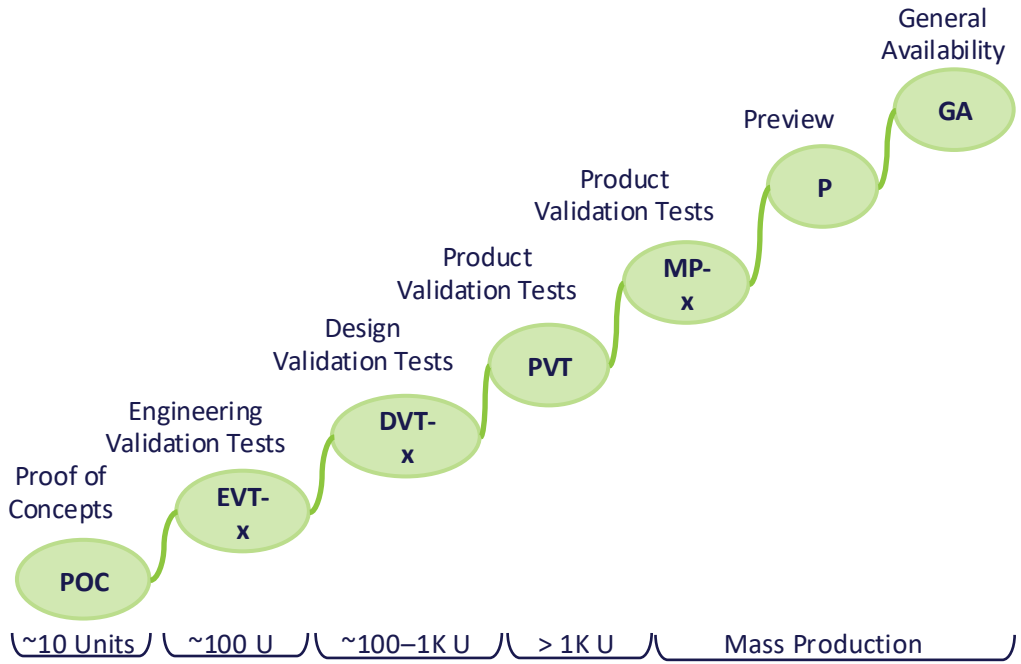
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The Problem: Debug at Hyperscale

- Cloud + AI scale the reliability math — even low DPPM means meaningful fleet-level failures
- Traditional JTAG and probe-based debug need physical access — impossible across a live fleet
- Root cause now needs hardware/software co-debug across CPUs, accelerators, firmware, OS, platform
- No standard, secure way to debug in-field across the lifecycle — without exposing customer workloads



Failure rate vs. time — the fleet lives in the flat middle



Where Existing Efforts Fall Short

Today: Management solutions in the data center and debugging in the lab for one blade.

Each layer is well solved in isolation. What is missing is the connective tissue between them — orchestration and execution across layers.

TODAY: Primarily Point Solutions

Silicon Debug
TRACE32® / GTL workflows, OpenOCD, etc.

Platform Management
e.g., Redfish, PLDM, MCTP, SPDM

Device Management

Host Debug
Serial-over-LAN, KVM-IP, etc.

Fleet Diagnostics

Vendor Solutions

THE SEAM TO STANDARDIZE — IN OCP®

Orchestration + execution, not just data & formats
OCP® System Debug Specification

Full-system, full-lifecycle coverage
OCP® Hardware Management • System Debug Workstream

BMC as debug gateway — no probes
OCP® DC-SCM™ • OpenBMC®

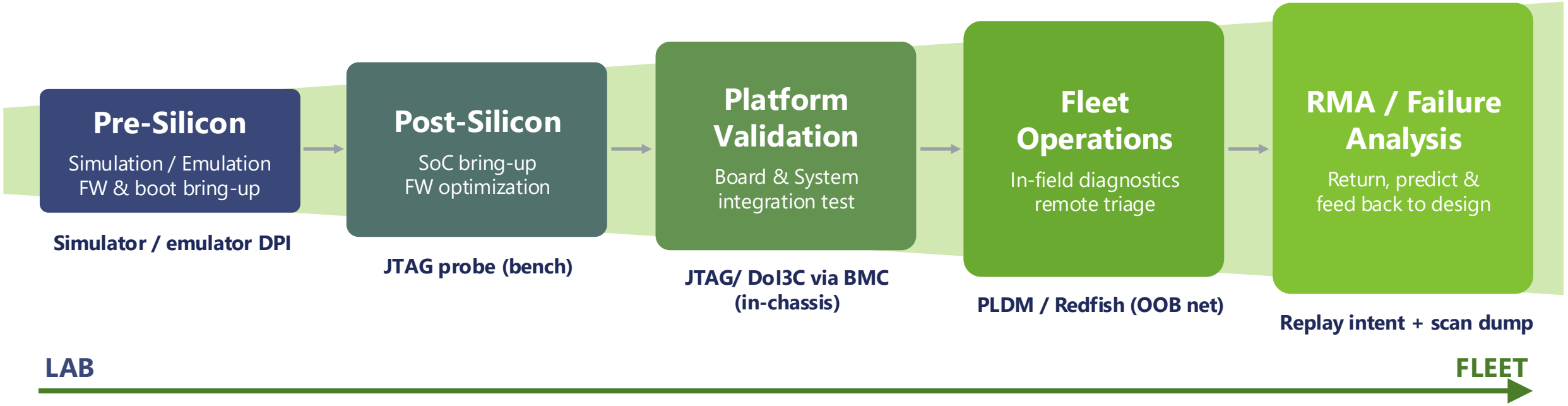
Secure multi-tenant scale
DMTF® SPDM® — authentication + authorization

Vendor-neutral: CPUs + accelerators
MIPI® SPP™ • Debug over I3C®

From Lab to Fleet

One framework, one debug intent — from pre-silicon simulation to in-field fleet diagnostics

Same debug intent at every stage — only the transport binding changes



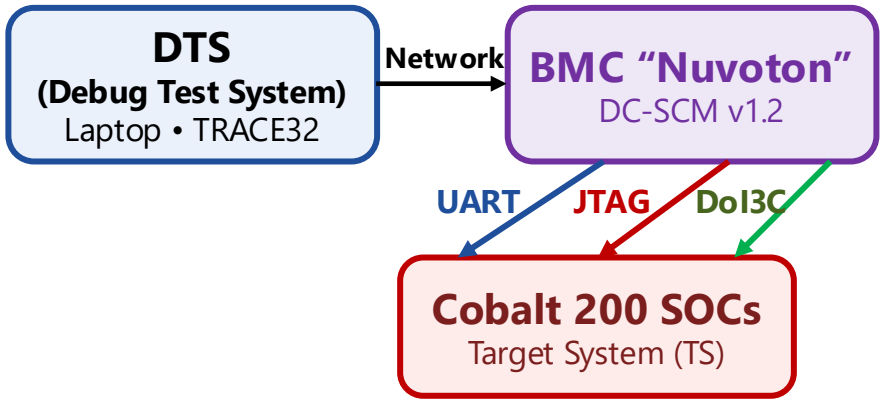
Standards-Based Transport • BMC Debug Gateway • Local Execution • Secure Multi-Tenant Access

Prototype Platform: Microsoft Cobalt 200

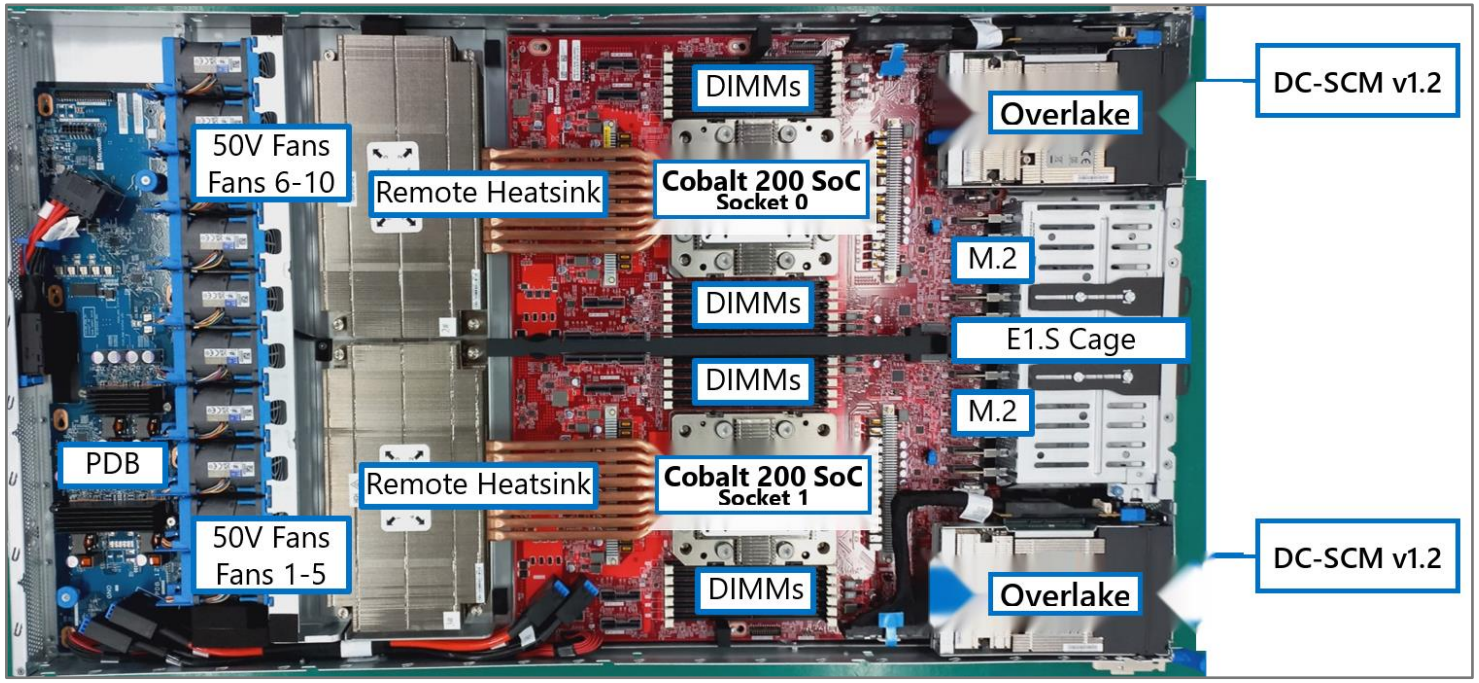
The live C200 node — how the DTS reaches the SoCs for in-chassis platform debug

Cobalt 200 - Platform Debug Setup

1. Debug Terminal (DTS) connected
2. to **DC-SCM v1.2** (Datacenter Secure Control Module) containing the **BMC "Nuvoton"**
3. PDB provides **Power** to Cobalt 200, Overlake SoCs and to the DC-SCM
4. **Cobalt 200 SoCs** (2 central packages) as the **Target System (TS)** — debugged in-chassis over **network** via the BMC



C200 Host Processor Module— Top view, Lid Removed



* Platform used to capture the **interactive-debug latency benchmarks**

On the live node: **Cobalt 200 SoC** = 2 central compute packages · **PDB / 50V bus** = platform power · **DC-SCM v1.2** = BMC
Debug Link: **DTS** → **BMC** (network) → **JTAG / MIPI DoI3C** (SoC scan chain) **/ UART** — all in-chassis

Latency Kills Interactive Debug

Internet round-trips — not encryption — destroy interactivity; local execution restores it

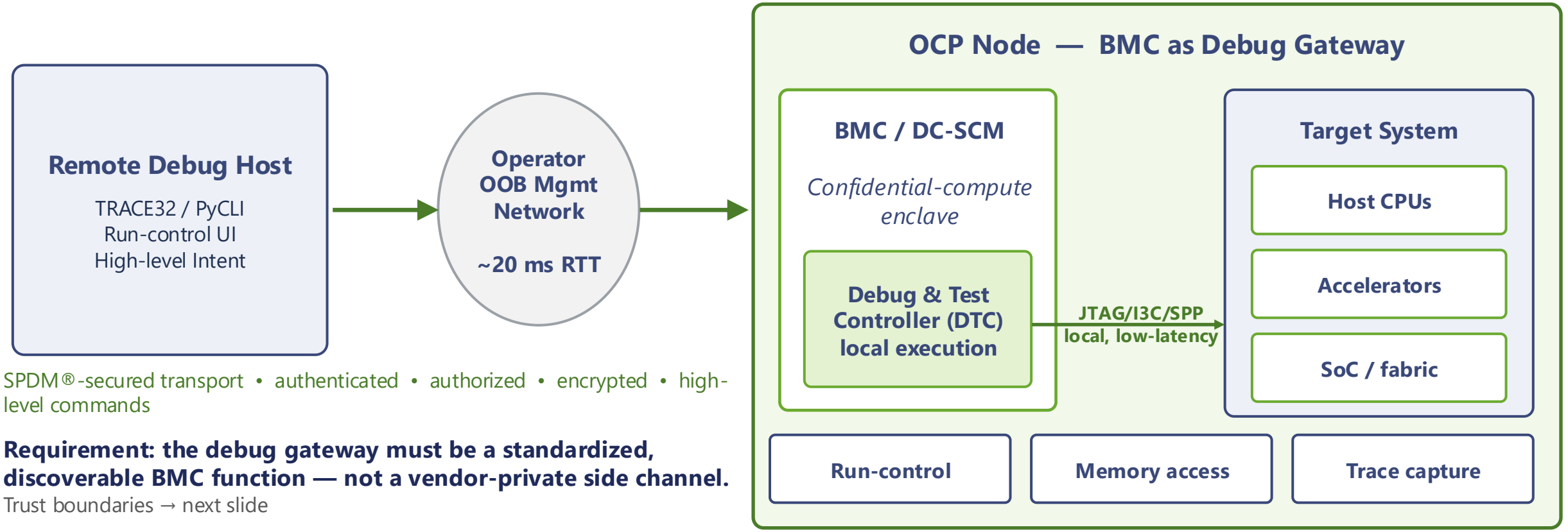
Benchmarks measured on
Microsoft® Cobalt 200 · prototype platform



BMC as Debug Gateway

The BMC performs run-control, memory access, and trace locally — no external probes

These benefits are most visible in heterogeneous architectures (e.g., Arm® + RISC-V domains observed concurrently)



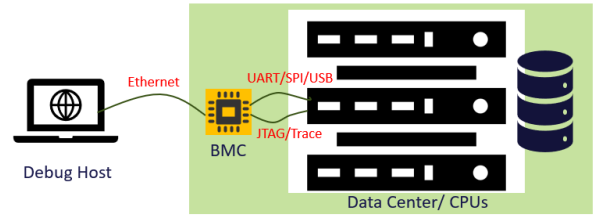
SPDM®-secured transport • authenticated • authorized • encrypted • high-level commands

Requirement: the debug gateway must be a standardized, discoverable BMC function — not a vendor-private side channel.

Trust boundaries → next slide

Secure, Multi-Tenant Debug Modes

SPDM® authentication proves identity; SPDM® authorization scopes the debug stream

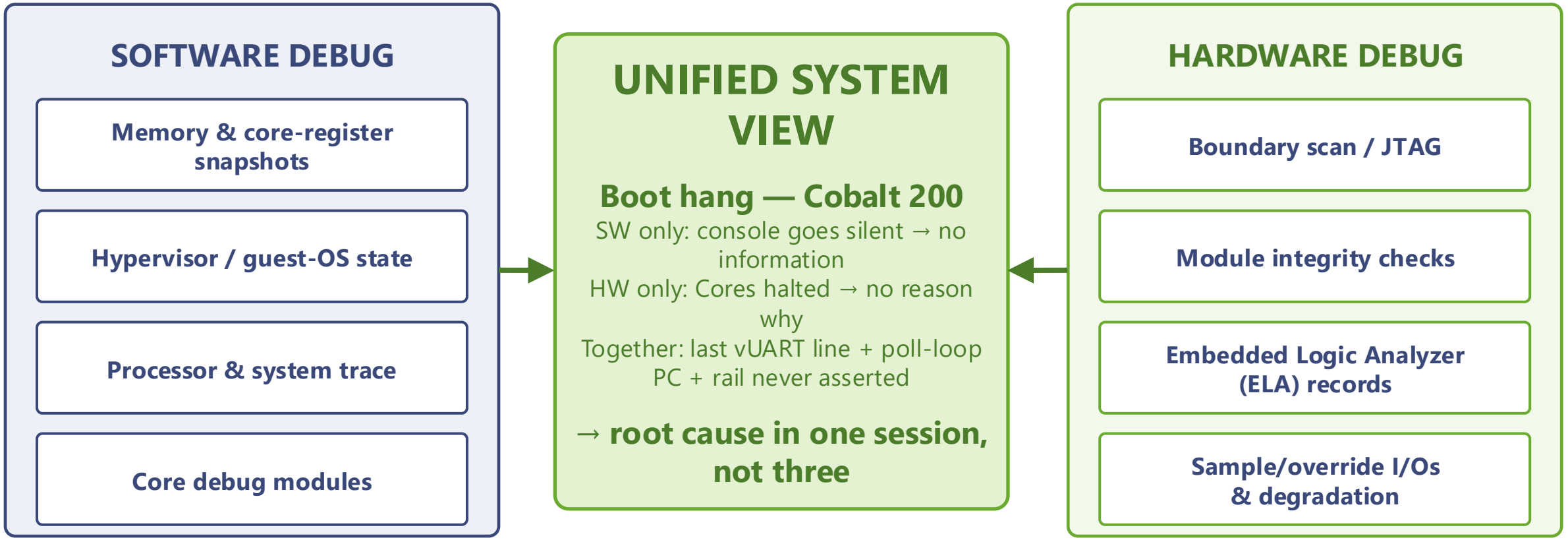


Four modes, one debug intent — target is a BMC confidential-compute (CC) gateway

#1 Non-Secure	#2 Quasi-Secure	#3 Endpoint-Secure	#4 Secure Gateway
Trusted lab. Complete setup in a trusted environment. **Trusted**	Authenticated + encrypted host→BMC session. Gateway in trusted BMC. Weaker trust model. **BMC trusted**	End-to-end host→target, Secure today but WAN latency hurts. **BMC untrusted**	Gateway in BMC confidential compute. Secure + local speed. **BMC untrusted + CC**
Perf: Highest	**Perf: High**	**Perf: Lowest**	**Perf: High**

Hardware / Software Co-Debug

Correlate software symptoms with hardware evidence for faster root cause



Vendor-Neutral & Standards-Based

MIPI transport, DMTF SPDM security, OCP / OpenBMC platform — extends to CPUs and accelerators

Vendor-neutral architecture built on open standards

Debug Applications	Run-Control • Memory Access • Trace • Co-Debug Orchestration
Security — DMTF SPDM®	Authentication • Authorization (debug-stream scope) • Encrypted Sessions • Multi-Tenant Access Control
Transport — MIPI Alliance	MIPI SPP™ • Debug over I3C® • Debug Security • Connector Rec.
Platform — OCP®	DC-SCM • OpenBMC • BMC as debug gateway / DTC

Extends beyond the BMC to host CPUs and accelerators • reusable for telecom, Edge AI, ADAS

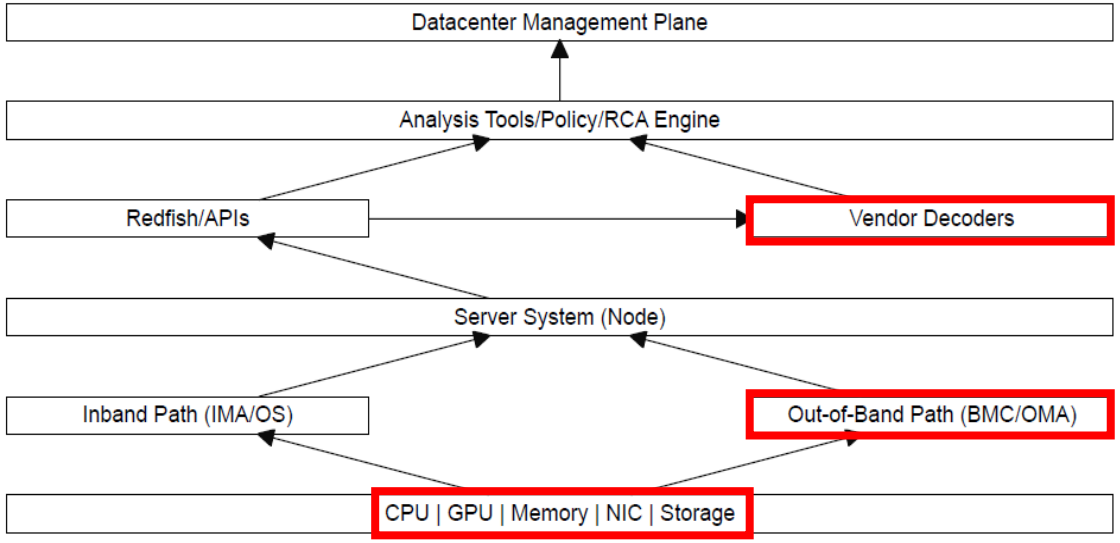
OCP: System Debug Specification

The goal is to define the use of common connectivity, transport, and protocols:

- Define a common communication “pipe” for debug data/messages.
- Data/message payloads stay vendor-specific.
- Allowing flexibility but providing interoperability.
  - Flexibility in how the system is put together.
  - Debug capabilities need to be discoverable and self-describing
  - Interoperability with both different components and different debug tooling.
  - Data needs to be opaque to components in the middle (e.g., BMC).

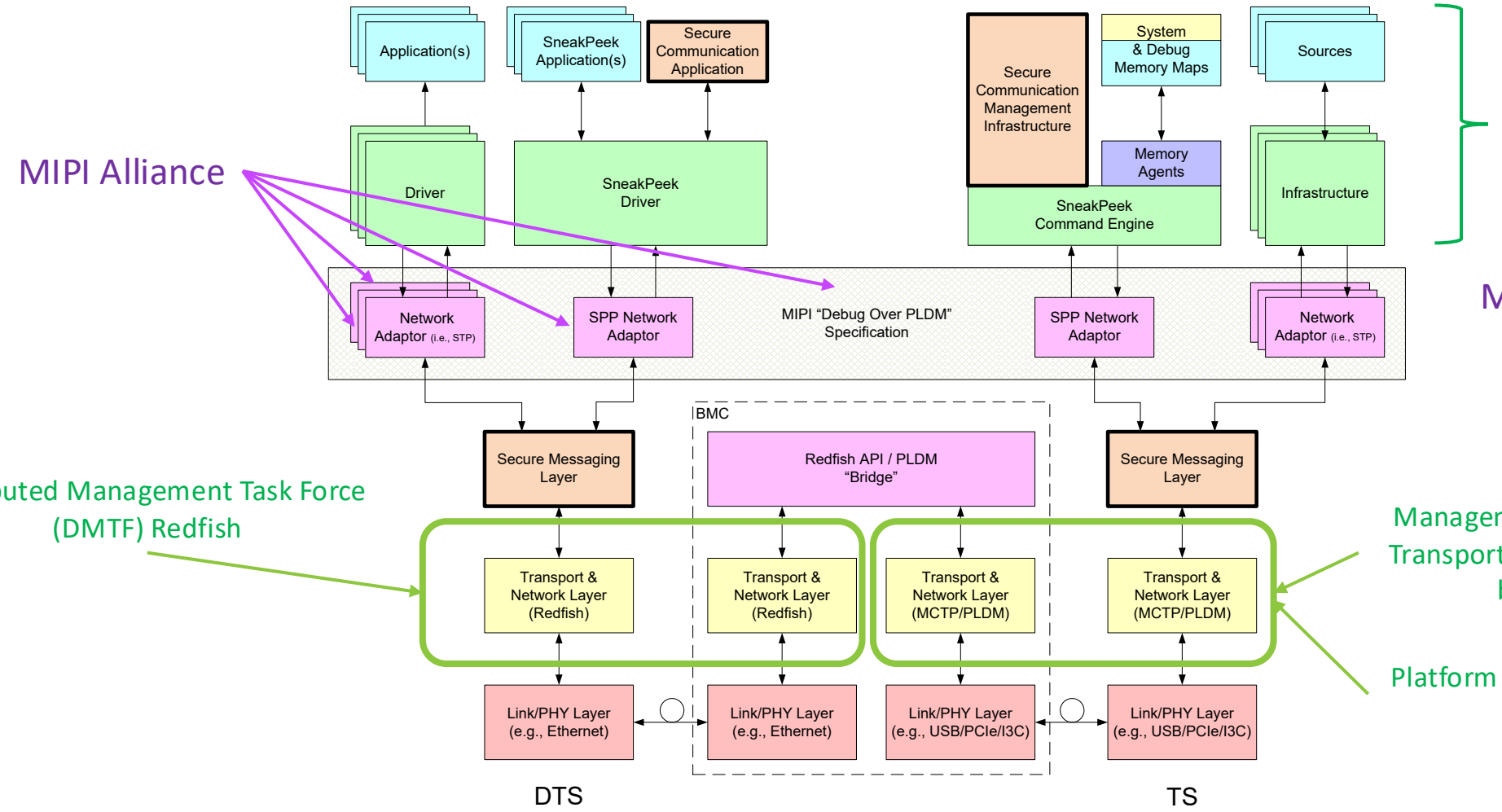
Out of Scope (covered by other standards):

- Error Events
- Bulk Dumping
- Crash Dumping
- Register Dumping
- Array Dumping



* Open Compute Project · System Debug Specification v0.1

MIPI Debug over PLDM (Conceptual)



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Key Takeaway

- Data-center reliability is mission-critical; triage needs full-system visibility
- One modular, secure debug framework spans the entire product lifecycle — lab to fleet
- BMC-as-gateway makes secure remote debug interactive — local execution beats the WAN
- SPDM® delivers authenticated, authorized, encrypted multi-tenant debug access
- Standards-based (MIPI®, OCP®, DMTF®) — vendor-neutral and scalable to adopt

→ **What we need from this room is on the next slide**

Call to Action — Where We Need Your Help

1. Join the OCP® Hardware Management — System Debug Workstream

Help refine the use cases, the security/trust boundaries, and the System Debug Specification.

opencompute.org/community/hardware-management · **ocp-all.groups.io/g/ocp-system-debug**

OCP Contacts: encrio.carrieri@ocproject.net, marko.bartscherer@ocproject.net

2. Join the MIPI® Alliance Debug Working Group

Transport and connectivity: SPP™, Debug over I3C®, and the debug security spec.

mipi.org/groups/debug

3. Engage DMTF® on SPDM authorization for debug streams

Authorization — not just authentication — makes multi-tenant in-field debug tractable.

dmtf.org/standards/spdm

4. Build and measure with us

Prototype on DC-SCM™ + OpenBMC®, bring a platform to interop, publish your latency numbers.

sduthiraru@microsoft.com · rolf.kuehnis@lauterbach.com

One action today: subscribe to ocp-system-debug and bring one real debug failure you could not root-cause remotely.



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Thank You!

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Questions & Discussion



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Backup — Supporting Detail

Pulled out of the main flow to keep the talk inside 20 minutes — happy to go to any of these in Q&A

- So What? Latency Becomes Downtime
- Execution flow — OCP® System Debug Specification v0.1
- Outlook: Cost Reduction
- References and prior art

So What? Latency Becomes Downtime

Interactive debug is not slow at 5 seconds per step — it is abandoned

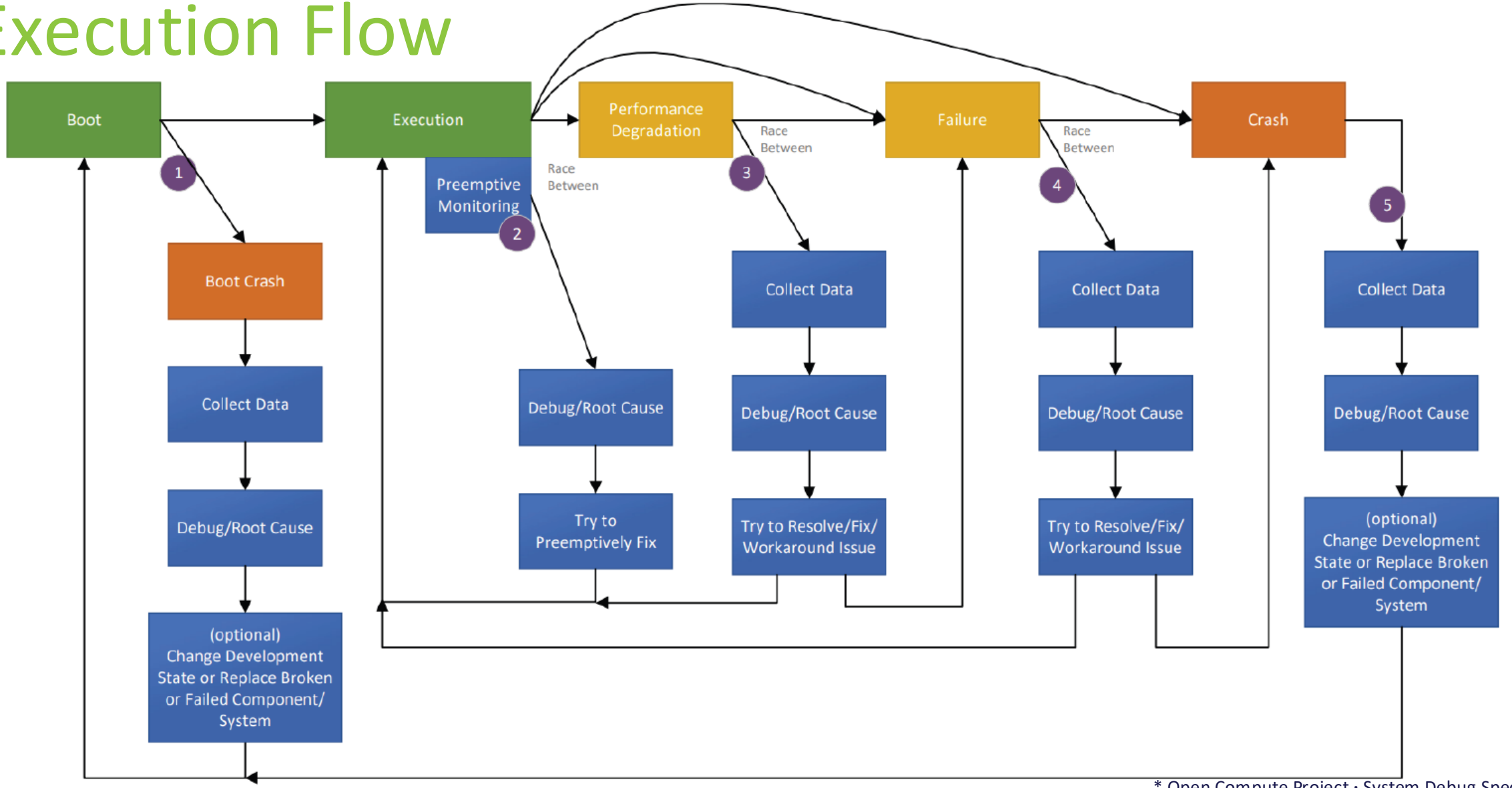
From milliseconds to Mean-Time-To-Root Cause (MTTR)

A REAL SESSION	THE BEHAVIORAL CLIFF	THE FLEET COST
500–2,000 single-steps per session Remote: 40 min – 2.7 h of stepping Local DTC: 12 – 50 seconds ~190× faster interactive debug	At ~5s per step, nobody steps. Engineers reboot-and-hope, or RMA. The cost is not slow debug — it is no debug at all.	No-Fault-Found returns and unexplained node retirements. A node under debug is out of rotation. Measure node-hours returned and MTTR.

Latency does not slow down debug — it eliminates it

Heterogeneous SoCs amplify this — ARM + RISC-V domains observed concurrently multiply round-trips per fault.

Execution Flow



* Open Compute Project · System Debug Specification v0.1

Outlook: Cost Reduction

- All Software Solution/ “Self-Hosted”
  - ~Zero BOM cost; Scalable
  - … but limited debug access and challenging when the CPU becomes unresponsive
- Low-cost, discrete BMC
  - Linear approach provides moderate savings by removing significant functionality (less compute power, smaller memory footprint, reduced interface capabilities, slower speed)
- Integration of BMC functionality into other components
  - Ideally will be always-on and provide out-of-band network access. Options:
    - Smart NIC/DPU
    - Baseboard PMIC/ Host Processor Module (HPM) CPLD/ System Control Processor (SCP) or similar discrete board component with a µController

Approach	BOM	Hard hang?	Pre-boot?
Self-hosted (CD/kdump/APEI)	~$0	✗	✗
Low-cost RTOS BMC	~$3–8	✓	✓ limited
Full BMC (DC-SCM v1.2)	~$8–20	✓	✓
Integrated (DPU / SCP / CPLD)	shared	✓	✓

Existing examples today

Self-hosted: kdump / kexec, ACPI/APEI + CPER, autonomous crash dump — all ~zero BOM, all need a live core.
Discrete BMC: Nuvoton on DC-SCM v1.2 — the very BMC on this prototype.
Integrated: DPU-hosted management (always-on, own memory, OOB network, root of trust); in-SoC management core (SCP) at zero incremental BOM.
Trade-off is real: cost falls, and the debug capability floor falls with it.

References

- AMI MegaRAC — ami.com/products/megarac/#onetree
- ASSET ScanWorks Embedded Diagnostics(SED) — asset-intertech.com/resources/blog/2017/02/at-scale-debug/
- DMTF SPDM (DSP0274) — dmtf.org
- Intel At-Scale Debug — github.com/Intel-BMC/asd/blob/master/README.md
- Lattice HPM CPLD — Lattice OCP Ready Solutions for DC-SCM and HPM CPLD Connectivity | Low-Power FPGA
- Lauterbach — GTL Plugin to Debug Servers via I3C or JTAG Controlled by a Board Management Controller
- Lauterbach GTL API / TRACE32 — lauterbach.com/products/software/debugger-for-simulators-emulators-and-virtual-targets/gtl
- MIPI Architecture Overview for Debug — mipi.org
- MIPI Security Specification for Debug — Security Specification for Debug | MIPI
- Multicore Debug API — https://www.lauterbach.com/products/software/debugger-for-simulators-emulators-and-virtual-targets/mcd-api
- OCP DC-SCM Specification — opencompute.org
- OCP Hardware Fault Management — https://github.com/opencomputeproject/OCP-HM-HFM-Hardware_Fault_Management